

Inés Ortega-Fernández

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Academic profile

Inés Ortega-Fernández is a researcher in machine learning, cybersecurity, trustworthy artificial intelligence, and privacy-preserving technologies. Her research interests focus on interpretable and trustworthy machine learning, explainable deep learning, anomaly detection, privacy-preserving AI, federated learning security, and applied computational statistics.

She received her PhD in Information and Telecommunications Technologies from the Universidade de Vigo in 2024, awarded with *Cum Laude* distinction, International Mention, and Industrial Doctorate distinction, with the dissertation entitled “Machine Learning Approaches and Explainability for Real-Time Cyberattack Detection”. During her doctoral studies, she carried out a predoctoral research stay at the Department of Computer, Control and Management Engineering (DIAG) of Sapienza University of Rome, where she investigated reinforcement-learning approaches for the mitigation of denial-of-service attacks in smart grids.

From 2022 to May 2026, she served as Technical Manager of Data Analytics & AI at the Galician Research and Development Center in Advanced Telecommunications (Gradiant), leading multidisciplinary R&D activities in artificial intelligence, cybersecurity, privacy-enhancing technologies, and data analytics. Since Summer 2026, she has been a Research Fellow in the highly competitive MATS Program (ML Alignment & Theory Scholars), currently in its extension phase, where she works on LLM inference verification (the statistical detection of model-weight exfiltration through steganographic covert channels in the token stream) under the mentorship of Keri Warr (Anthropic). Previously, she worked as Software Engineer at the Microsoft Canada Development Centre within the Core Data Engineering group, developing large-scale telemetry processing solutions for Windows and Azure cloud infrastructures. Earlier in her career, she also worked as research assistant at Universidad Carlos III de Madrid on biometric data-processing and privacy-related research projects. She additionally held a position as Associate Lecturer at the Department of Statistics and Operations Research at the Universidade de Vigo during 2023/2024 and 2025/2026 academic years.

Her research focuses on trustworthy and secure artificial intelligence, particularly in the areas of AI safety, privacy-preserving machine learning, explainability, and cybersecurity. She has worked extensively on anomaly and cyberattack detection for industrial and critical infrastructures, behavioral analytics (UEBA), privacy attacks and defenses in federated learning, differential privacy, and anonymization methods for sensitive data. Her work also explores interpretable and robust machine learning systems, combining modern deep learning approaches with principles from statistical modeling such as Generalized Additive Models to achieve transparency and trustworthy AI.

She strongly believes in open, transparent, and reproducible science. As part of this commitment, she has developed and maintained open-source scientific software, most notably the **neuralGAM** packages for R and Python, implementing interpretable neural generalized additive models. These tools have accumulated more than 250,000 downloads worldwide and are used for interpretable machine learning and statistical modeling applications.

During the last years, she has participated in 14 competitive European, national, and regional R&D projects, including Horizon Europe, H2020, CDTI Cervera networks, INCIBE, and Red.es initiatives, with a combined budget exceeding 60 million euros. She has acted as principal investigator, consortium coordinator, work-package leader, technical lead, and research team member in projects related to cybersecurity, privacy-preserving AI, industrial anomaly detection, big-data infrastructures, and trustworthy machine learning systems.

Her scientific contributions include peer-reviewed publications in journals such as *Information Systems Frontiers*, *Statistics and Computing*, *Wireless Networks*, *AIMS Mathematics*, or *The R Journal*, as well as contributions to international conferences and workshops including IEEE WIFS, ACM IH&MMSec, IEEE EuroS&PW, and JNIC. Her work has received more than 170 citations in WoS and Scopus, with a field-weighted citation impact of 1.60 (above world average). In addition, she has supervised bachelor, master, and doctoral theses in cybersecurity, privacy-preserving AI, and explainable machine learning, including award-winning student projects.

Besides her research activities, she actively contributes to the scientific community as reviewer and program committee member. She has served on the Scientific Program Committee of the IH&MMSec (2024), the Spanish National Cybersecurity Research Conference (JNIC) (2023–2025), and was recognized as Gold Reviewer at the International Conference on Machine Learning (ICML 2026). She has also served as reviewer

for different peer-review journals, including the International Journal of Machine Learning and Cybernetics, Artificial Intelligence Review, and Peer-to-Peer Networking and Applications.

Her research activity combines machine learning, cybersecurity, telecommunications, privacy-enhancing technologies, and computational statistics in close collaboration with multidisciplinary teams from academia, industry, and public institutions. She has collaborated with companies and organizations such as the University of Vigo, Cellnex, ABANCA, INCIBE, CCN-CERT, and several European research consortia, contributing to the transfer of trustworthy AI and cybersecurity technologies into real-world applications. She has also participated in science communication and citizen-science initiatives, including the development of the collaborative environmental monitoring platform `pelletMap` during the 2023–2024 Galician pellet spill crisis.

More information on <https://inesortega.github.io>

Education

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| 2020–2024 | PhD in Information and Telecommunications Technologies , University of Vigo, Spain. Thesis: <i>Machine Learning Approaches and Explainability for Real-Time Cyber-attack Detection</i> . Advisors: Marta Sestelo and Juan C. Burguillo. Committee: Bipin Indurkha (Jagiellonian University), Ana Fernández-Vilas (University of Vigo), Alejandra Cabaña (Universidad Autónoma de Barcelona). Grade: <i>Cum Laude</i> . Research focus: interpretable machine learning, anomaly detection, cybersecurity, and real-time cyberattack detection. |
| 2017–2018 | MSc in Cybersecurity , Carlos III University of Madrid, Spain. Focus on secure systems, network defense, vulnerability analysis, cryptography, and cyber-risk assessment. |
| 2012–2017 | BSc in Computer Science , Carlos III University of Madrid, Spain. Minor in Computer Engineering. Exchange year at Aalto University, Finland, 2015–2016. |

Employment

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| 2026–present | Research Fellow , MATS Program (ML Alignment & Theory Scholars). Summer 2026 cohort, followed by the extension phase (remote, through early 2027). Research on LLM inference verification and the detection of model-weight exfiltration under the mentorship of Keri Warr (Anthropic) and Lisa Thiergart (Security Level 5). |
| 2022–2026 | Technical Manager of Data Analytics & AI , Security and Privacy Department, Gradiant, Vigo, Spain. Lead Scientist of the Data Analytics & AI team at the Security and Privacy Department. Management and development of R&D projects applying machine learning and AI to cybersecurity, data privacy, privacy-preserving AI, and trustworthy systems. |
| 2023–2024
2025–2026 | Lecturer , Department of Statistics and Operations Research, University of Vigo, Spain. Teaching the course “Statistics” for the BSc in Business Administration, including descriptive statistics, probability, random variables, and parametric inference. |
| 2020–2022 | Senior Researcher / Engineer , Gradiant, Vigo, Spain. Research and development of machine learning methods for security in industrial environments, including unsupervised anomaly detection, clustering, anonymization, and privacy-preserving techniques. |
| 2018–2020 | Software Engineer , Microsoft, Vancouver, Canada. Development of big-data infrastructure and telemetry-processing workflows for Windows systems. |

Research stays

Predoctoral Research stays

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| 2023 | Department of Computer, Control and Management Engineering, Sapienza University of Rome, Italy. Francesco Liberati. Sept-Dec 2023. |
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Honors, awards and grants

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| 2026 | MATS Research Fellowship , ML Alignment & Theory Scholars Program. Competitive international fellowship in AI safety and alignment research. |
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- 2026 **Gold Reviewer** (top 25%) at the International Conference on Machine Learning (ICML 2026).
- 2026 **Best PhD Thesis Award**, USC-INCIBE Cybersecurity Economics Chair (CECOCIB).
- 2025 **Best PhD Thesis Award in Cybersecurity**, Cátedra R en Ciberseguridad (UVigo – UDC).

Publications

Journal articles

- 2026 Padín-Torrente, H., Carneiro-Díaz, V., & Ortega-Fernandez, I. (2026). Toward human-centered explainability: Natural language explanations for anomaly detection. *Information Systems Frontiers*, 1–17. <https://doi.org/10.1007/s10796-026-10717-3>
- 2026 Ortega-Fernandez, I., & Sestelo, M. (2026). neuralGAM: An R package for fitting generalized additive neural networks. *The R Journal*, 18, 253–276. <https://doi.org/10.32614/RJ-2026-016>
- 2025 Fuentes, J., Ortega-Fernandez, I., Villanueva, N. M., & Sestelo, M. (2025). Cybersecurity threat detection based on a UEBA framework using deep autoencoders. *AIMS Mathematics*, 10(10), 23496–23517. <https://doi.org/10.3934/math.20251043>
- 2024 Ortega-Fernandez, I., Sestelo, M., Burguillo, J. C., & Piñón-Blanco, C. (2024). Network intrusion detection system for DDoS attacks in ICS using deep autoencoders. *Wireless Networks*, 30(6), 5059–5075. <https://doi.org/10.1007/s11276-022-03214-3>
- 2024 Ortega-Fernandez, I., Sestelo, M., & Villanueva, N. M. (2024). Explainable generalized additive neural networks with independent neural network training. *Statistics and Computing*, 34(1), 6. <https://doi.org/10.1007/s11222-023-10320-5>
- 2023 Ortega-Fernandez, I., & Liberati, F. (2023). A review of denial of service attack and mitigation in the smart grid using reinforcement learning. *Energies*, 16(2). <https://doi.org/10.3390/en16020635>
- 2019 Sanchez-Reillo, R., Ortega-Fernandez, I., Ponce-Hernandez, W., & Quiros-Sandoval, H. C. (2019). How to implement eu data protection regulation for r&d in biometrics. *Computer Standards & Interfaces*, 61, 89–96. <https://doi.org/10.1016/j.csi.2018.01.007>

Conference and workshop papers

- 2026 Ortega-Fernandez, I., Kowalczyk, M., & Warr, K. (in press). A conformal layer for fixed-seed LLM inference verification. *2026 IEEE International Workshop on Information Forensics and Security (WIFS)*. Oral presentation; Sendai, Japan, December 7–11, 2026
- 2026 Ortega-Fernandez, I., Kowalczyk, M., & Warr, K. (2026). Anytime-valid detection of LLM weight exfiltration. *E-Values: From Statistics to ML, NeurIPS 2026 Workshop*. Oral presentation (one of four contributed talks); Paris, France, December 13, 2026; non-archival
- 2026 Kryza, B., Ślota, R., Dutka, Ł., Kitowski, J., Aznar-Poveda, J., Fahringer, T., Ortega-Fernandez, I., Abad, I., Dlugolinsky, S., Bobak, M., Hluchy, L., Škrlec, I., & Kišša, P. (2026). Data management and i/o provisioning across cloud-edge continuum for high-performance computational data pipelines. In M. Paszynski, A. S. Barnard, & Y. J. Zhang (Eds.), *Computational Science – ICCS 2026 Workshops* (pp. 535–550). Springer Nature Switzerland
- 2026 Padín-Torrente, H., & Ortega-Fernandez, I. (2026). On the Practical Viability of Local Agentic Language Models for Android Security Analysis [Poster presentation]. *Jornadas Nacionales de Investigación en Ciberseguridad (JNIC 2026)*

- 2024 Loureiro-Acuña, J., Martínez-Luaña, X., Padín-Torrente, H., Jiménez-Balsa, G., García-Pagán, C., & Ortega-Fernandez, I. (2024). Enhancing Privacy in Federated Learning: A Practical Assessment of Combined PETs in a Cross-Silo Setting. *Proceedings of the 2024 ACM Workshop on Information Hiding and Multimedia Security*, 265–270. <https://doi.org/10.1145/3658664.3659661>
- 2024 Carnero Ortega, D., Piñón Blanco, C., Pintos Castro, B., & Ortega-Fernandez, I. (2024). Development of a modular virtual Industrial Control System prototype for cybersecurity research. *Jornadas Nacionales de Investigación en Ciberseguridad (JNIC) (9ª.2024. Sevilla) (2024)*, pp. 534-539., 534–539. <https://hdl.handle.net/11441/159179>
- 2024 Pintos Castro, B., & Ortega-Fernandez, I. (2024). What the file, análisis forense basado en inteligencia artificial. *Jornadas Nacionales de Investigación en Ciberseguridad (JNIC) (9ª.2024. Sevilla) (2024)*, pp. 534-539., 600–605. <https://hdl.handle.net/11441/159179>
- 2023 Rodriguez-Viñas, J., Ortega-Fernandez, I., & Martínez, E. S. (2023). Hexanonymity: A scalable geo-positioned data clustering algorithm for anonymisation purposes. *2023 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 396–404. <https://doi.org/10.1109/EuroSPW59978.2023.00050>
- 2023 Piñón-Blanco, C., Otero-Vázquez, F., Ortega-Fernandez, I., & Sestelo, M. (2023). Detecting Anomalies in Industrial Control Systems with LSTM Neural Networks and UEBA. *2023 JNIC Cybersecurity Conference (JNIC)*, 1–8. <https://doi.org/10.23919/JNIC58574.2023.10205609>
- 2023 Saavedra Golán, M., & Ortega-Fernandez, I. (2023). Detección de “bots” avanzados en comercio electrónico: Un caso de uso real. *Actas de las VIII Jornadas Nacionales de Investigación en Ciberseguridad: Vigo, 21 a 23 de junio de 2023*, 229–235. <https://hdl.handle.net/11093/4952>
- 2017 Reillo, R. S., Ortega-Fernandez, I., Hernandez, W. P., & Sandoval, H. Q. (2017). How to Implement EU Data Protection Regulation for R&D on Personal Data. *Proceedings of the 2017 International Carnahan Conference on Security Technology (ICCST)*. <https://doi.org/10.1109/CCST.2017.8167797>

Book chapters

- 2022 Ortega-Fernandez, I., Martinez, S. E. K., & Orellana, L. A. (2022). Large Scale Data Anonymisation for GDPR Compliance. In J. Soldatos & D. Kyriazis (Eds.), *Big data and artificial intelligence in digital finance: Increasing personalization and trust in digital finance using big data and ai* (pp. 325–335). Springer International Publishing. https://doi.org/10.1007/978-3-030-94590-9_19

Conference presentations

- 2025 Ortega-Fernández, I., Sestelo, M., & Villanueva, N. M. (2025). Towards Explainable AI: Neural Network-Based Training of Generalized Additive Models [Reviewed contribution accepted for conference presentation, December 13–15, 2025]. *Royal Statistical Society International Conference (RSS 2025)*
- 2025 Fuentes Rodríguez, J., Ortega-Fernandez, I., Villanueva, N. M., & Sestelo, M. (2025). Advanced Detection of Suspicious Activity within UEBA Framework using Deep Autoencoders [Oral communication]. *XVII Congreso Galego de Estatística e Investigación de Operacións and II Xornadas de Innovación Docente na Estatística e Investigación de Operacións*
- 2025 Ortega-Fernandez, I., Sestelo, M., & Villanueva, N. M. (2025). Combining Neural Networks and Generalized Additive Models for Explainable AI [Reviewed contribution accepted for poster presentation]. *XVII Congreso Galego de Estatística e Investigación de Operacións and II Xornadas de Innovación Docente na Estatística e Investigación de Operacións*. Poster presentation

- 2025 Makri, F., Spantideas, S., Kokkinis, G., Ortega-Fernandez, I., Alonso Doval, P., & Varveris, M. (2025). The future of law enforcement: How PRESERVE's AI and big data solutions benefit public safety [Reviewed contribution accepted for poster presentation]. *2025 6th International Conference in Electronic Engineering & Information Technology (EEITE)*
- 2024 Ortega-Fernandez, I., & Sestelo, M. (2024). Explainable deep learning: A methodology to train Generalized Additive Models with deep neural networks [Reviewed contribution accepted for poster presentation]. *International Symposium on Nonparametric Statistics (ISNPS 2024)*
- 2023 Ortega-Fernández, I., & Sestelo, M. (2023). Explainable Generalized Additive Neural Networks with Independent Neural Network Training [Reviewed contribution accepted for poster presentation]. *16th International Conference of the ERCIM WG on Computational and methodological Statistics (CFE-CMStatistics 2023)*

Preprints

- 2025 Montaña-Fernández, P., & Ortega-Fernandez, I. (2025). An efficient gradient-based inference attack for federated learning. *arXiv preprint arXiv:2512.15143*

Invited research seminars and workshops

- 2026 “How leaky is your inference API? Bounding and detecting steganographic exfiltration in the token stream”. AI Security Forum. Las Vegas, NV, USA. August 6, 2026.
- 2026 “From Faster Incident Response to New Security & Privacy Risks”. AI & Humanity Research Hub Workshop on “Securing the Future of AI: From Regulations to Neuromorphic Frontiers”, February 24th, 2026, Loughborough University, London, UK.
- 2025 “How Private is an AI Model? Understanding Attacks, Metrics & PETs”. Session: “Privacy Preserving Technologies (PPT) in Sharing of Personal Data for AI Development and Regulatory Compliance” at the 4th Privacy Symposium, May 12–16, 2025, Venice, Italy
- 2023 H2020 INFINITECH Stakeholders’ Webinar (On-Line) “Data Management: Infinittech case-studies”. February 23rd, 2023.
- 2022 “Detectando ciberataques y comportamientos anómalos mediante UEBA: un caso de uso real en Abanca”. XVI Jornadas STIC CCN-CERT and IV Jornadas de Ciberdefensa ESPDEF-CERT, December 2, 2022, Madrid, Spain.
- 2022 “Data anonymization: becoming GDPR compliant in the era of AI and Big Data”. Invited talk at the “Women in technology behind data-sharing, privacy preservation and Self-Sovereign identity” from the KRAKEN EU Project. May 31st, 2022.

Software

- 2024 **neuralGAM**. Author of **neuralGAM** Neural Network framework based on Generalized Additive Models, which trains a different neural network to estimate the contribution of each feature to the response variable. The networks are trained independently leveraging the local scoring and backfitting algorithms to ensure that the Generalized Additive Model converges and it is additive. The resultant Neural Network is a highly accurate and interpretable deep learning model, which can be used for high-risk AI practices where decision-making should be based on accountable and interpretable algorithms.
- 2023 **pelletMap**. R/Shiny web application for real-time citizen monitoring of the 2023–2024 Galician plastic pellet spill, integrating automated data ingestion, geocoding, and interactive map visualization and statistical analysis.
- 2023 **Hexanonymity**. Scalable algorithm for anonymization of geo-positioned data based on clustering methods. Code available as open-source contribution: GitHub

Teaching

Teaching

2023–2024 **Lecturer**, Department of Statistics and Operations Research, University of Vigo. Teaching Statistics for the BSc in Business Administration, including descriptive statistics, probability, random variables, and parametric inference.
2025–2026

Thesis Advisor

2025–present **Explainable Artificial Intelligence for Anomaly Detection in Cybersecurity**. Student: Héctor Padín Torrente. Programme: Official PhD Programme in Information and Communication Technologies, Universidade da Coruña. Role: Co-supervisor (with Víctor Carneiro-Díaz, UDC).
2026 **Dynamic Differential Privacy for Improving the Privacy–Utility Trade-off in Federated Learning**. Student: Carlos García-Pagán García. MSc in Mathematics, UNED. Role: Co-supervisor (with Jorge M. Arevalillo, UNED).
2024 **Study and Implementation of Practical Privacy Attacks on Machine Learning in Federated Learning Settings**. Student: Roy Covelo Vázquez. BSc in Computer Engineering, University of Vigo. Grade: Honors. Awarded Best Bachelor’s Thesis in Cybersecurity. Role: Co-supervisor (with María Reyes Pavón Rial, UVigo).
2022 **Cyberattack Detection via User and Entity Behavior Analytics**. Student: Mauro Saavedra Golán. MSc in Big Data Analytics, Universidade de Santiago de Compostela. Grade: 9. Role: Co-supervisor (with Juan Carlos Vidal Aguiar, USC).
2022 **Implementation and Demonstration of Anonymization Algorithms for Geopositioned Data**. Student: Javier Rodríguez Viñas. BSc in Computer Engineering, Universidade de Santiago de Compostela. Grade: 9.3. Role: Co-supervisor (with Eduardo Sánchez Vila, USC).

Practices Advisor

2021 Supervisor of external academic practices (225 hours), BSc. on Computer Science (University of Santiago de Compostela). Student: Javier Rodríguez Viñas.

Patent and software registration

2026 **Software Registration** for “Cyber Behaviour Analysis Platform for Threat Detection”, Gradiant’s technology for UEBA-based cyberattack detection in SOC environments.

Research projects participation

2025–2026 **SPICE** – Smart Data Pipelines for the Cognitive Compute Continuum. Recovery and Resilience Plan of the Slovak Republic, GA 09I02-03-V01-00012. Role: Research team member. Research and development of privacy-preserving data anonymization solutions for integration within cloud-edge computing continuums and European data spaces.
2025–2026 **FAIR** – “Fight Fire with FAIR”. INCIBE (Spanish Cybersecurity Institute), CPP4 – CPP001/24. Role: Research team member. Explainability in neural networks for image classification tasks.
2024–2026 **PRESERVE** – Ethics and Privacy-preserving Big Data platform for Supporting criminal investigations. Horizon Europe, GA 101168309. Role: Principal Investigator and work-package leader. Research and development of privacy-preserving federated learning, UEBA, NLP, and computer-vision methods for law-enforcement contexts.
2023–2026 **SafeNet UEBA** – “Centro de Operaciones de Seguridad basado en UEBA explicable”. INCIBE (Spanish Cybersecurity Institute), CPP001-23-R006_SAFENET_UEBA. Role: Principal Investigator and consortium coordinator. Design and implementation of explainable AI and statistical methods for UEBA-based cyberattack detection in SOC environments.

2022–2026	SecBluRed – “Aproximación holística a la ciberseguridad en el IoT industrial”. CDTI Misiones Program, MIG-20221051. Role: Research team member. Development of AI-based cybersecurity solutions for Industrial IoT environments, including anomaly detection and intelligent monitoring.
2022–2025	TRUMPET – Trustworthy Multi-site Privacy Enhancing Technologies. Horizon Europe, GA 101070038. Role: Research team member. Research on privacy-enhancing technologies, differential privacy, and membership inference attacks in federated learning systems.
2023–2025	CICERO – “Contra medidas inteligentes de ciberseguridad para la red del futuro”. CDTI Cervera Network, CER-20231019. Role: Research team member. Investigation of privacy-preserving techniques and security mechanisms for large language models and future communication networks.
2021–2024	SmartNOC – “Investigación en tecnologías emergentes para la gestión inteligente de centros de control de redes de comunicación”. CDTI CIEN Program, IDI-20210861. Role: Research team member. Research on intelligent and secure management technologies for communication network operation centers and Industry 4.0 infrastructures.
2020–2023	BIECO – Building Trust in Ecosystems and Ecosystem Components. H2020, GA 952702. Role: Research team member. Research on vulnerability analysis, prediction, and propagation in ICT supply-chain ecosystems and complex infrastructures.
2020–2023	PERSIST – Patients-centered SurvivorShip care plan after Cancer treatments based on Big Data and AI technologies. H2020, GA 875406. Role: Research team member. Development of data anonymization and privacy-preserving techniques for healthcare and oncology data analytics.
2019–2022	INFINITECH – Tailored IoT and Big Data Sandboxes and Testbeds for Smart, Autonomous and Personalized Services in the European Finance and Insurance Ecosystem. H2020, GA 856632. Role: Task leader and research team member. Development of scalable anonymization algorithms for geo-positioned and sensitive data in finance and mobility contexts.
2019–2022	ÉGIDA – “Red de Excelencia en Tecnologías de Seguridad y Privacidad”. CDTI Cervera Network, CER-20191012. Role: Research team member. Research on explainable AI, cybersecurity analytics, privacy protection, and industrial intrusion detection systems.
2021–2022	SABOT – “Sistema AntiBOTS”. Red.es AI Program, 2020/0720/000100025. Role: Principal Investigator and scientific coordinator. Development of machine-learning and statistical methods for bot detection and behavioral analytics in e-commerce platforms.
2019–2022	CONGALS4.0 – “Diseño y Desarrollo de un modelo Industrial 4.0 en el Sector de Alimentación Gallego”. FEDER Industria 4.0 Program, IN854A 2019/15. Role: Research team member. Research on AI-based cyberattack detection methods for industrial and critical infrastructures.

Computer skills and languages

Programming Python, R, C, C++, Java, Bash, LaTeX.

Machine learning PyTorch, TensorFlow, JAX, scikit-learn, interpretable neural models, anomaly detection, federated learning, privacy-preserving machine learning.

Scientific computing Statistical modeling, simulation design, large-scale experimental evaluation, reproducible research.

Systems and tools Git, Linux, Docker, Kubernetes, SLURM, HPC clusters, distributed training.

Languages Spanish: native. Galician: native. English: professional working proficiency. Portuguese: intermediate. Italian: basic proficiency.

Service and professional activity

2025	MSc thesis committee member (MSc. on Information Security), University of Minho, Portugal. “ <i>Location Privacy Protection through Road Network Adaptability and User Mobility Prediction</i> ”. Student: Lara Miranda dos Santos.
2024	Member of the Scientific Program Committee of the ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec 2024).
2023–2025	Member of the Scientific Program Committee of the Spanish National Cybersecurity Research Conference (JNIC).
2022–Present	Reviewer for ICML, NeurIPS, AAAI, The R Journal, International Journal of Machine Learning and Cybernetics, Artificial Intelligence Review, and Peer-to-Peer Networking and Applications, among others.

Professional Training and Certifications

2025	Certificate of Achievement in <i>Fundamentals of LLMs (The LLM Course</i> , Hugging Face). Coursework on large language models, transformers, tokenization, attention mechanisms, and generative AI architectures.
2025	Training course “Las claves para la evaluación del desempeño” (CGADE Management School, 9 hours), focused on performance evaluation, feedback, communication, and leadership skills.
2023	Training course on equality and inclusion: “Sensibilización en materia de igualdad de oportunidades” (Igualia Formación, 4 hours).
2022	Training course “Liderazgo y Gestión de Equipos en Proyectos TI Ágiles” (Vitae, 10 hours), focused on agile leadership and management of technical teams in IT projects.
2021	Professional training in DevOps methodologies and cloud-native infrastructures (Asociación IDiA, 13.5 hours), including CI/CD, Kubernetes, Infrastructure as Code, observability, monitoring, and DevSecOps practices.
2021	Attendance at the VIII <i>Xornada de Usuarios de R en Galicia</i> , focused on statistical computing, data analysis, and open-source software development with R.

Additional training and certifications

2021	Scuba Diving, 1 Star, FEDAS.
2012	Celga 4, Galician Language Certificate (full proficiency).

Media Coverage and Public Outreach

2026	Featured in <i>COPE Galicia</i> and <i>La Voz de Galicia</i> for contributions to cybersecurity research and innovation in Galicia (CECOCIB Award for Best PhD thesis).
2024	The collaborative citizen-science platform pelletMap , developed with <i>Noia Limpa</i> during the Galician pellet spill crisis, received media coverage in <i>La Sexta</i> , <i>El Español – Quincemil</i> , <i>Nós Diario</i> , <i>Spanish Revolution</i> , and <i>Periodismo Ciudadano</i> .
2023	Interviewed by <i>Faro de Vigo</i> regarding the development of AI-based bot-detection systems for cybersecurity applications.
2021	Featured in <i>Atlántico Diario</i> as part of a report on women researchers and innovation in R&D+i sectors.